TEACHER GUIDE

Cambridge International AS & A Level
Business (9609)

# Investment appraisal

10 Finance and accounting (A Level)
**10.3 Investment appraisal**



A revision lesson. Pupils run a car dealership in firms of four and revise investment appraisal on their own deals. Seat each firm together. Each firm answers the topic tasks together.

## The lesson

**45-minute setting**

| Setup | Round 1 | Round 2 | Round 3 | Final questions |
|---|---|---|---|---|

### EVERY ROUND

**Firm decisions**
3 min

→

**Deals**
4½–5 min

→

**Topic tasks**
4 min

→

**B2B deal**
2½ min
from Round 2

### LESSON SETTINGS

| 30 min | **45 min** | 60 min | 75 min |
|---|---|---|---|

# What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



## CEO

**Staff · Sabotage** from Round 2



## Finance

**Capital & Loans · Investments**



## Sales & Marketing

**Marketing**



## Operations

**Seasonal stock**

# Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

## 1 Average profit

Round 1

New charging points.

**Average profit each year?**

| | |
|---|---|
| Cost to set up | **£30,000** |
| Returns over its life | **£66,000** |
| It lasts | **4 years** |

**ANSWER**

**£9,000**

(£66,000 − £30,000) ÷ 4 = £9,000

**Why:** Average yearly profit is total profit shared over the years.

## 2 Limits of payback

Round 1

You choose between two projects by payback.

**Which two are limits of payback?**

- **A** It ignores returns after payback
- **B** It is hard to work out
- **C** It ignores the time value of money
- **D** It favours slow projects
- **E** It ignores the project's cost

**ANSWER**

**✓ It ignores returns after payback**
**✓ It ignores the time value of money**

**Why:** Payback stops at the payback point and treats a pound later as a pound now.

## 3 Rank by payback

Round 2

Three projects with even yearly returns.

**Order by payback, fastest first?**

- **A** Cost £60,000, £20,000 a year
- **B** Cost £30,000, £12,000 a year
- **C** Cost £40,000, £10,000 a year

**ANSWER**

1. **Cost £30,000, £12,000 a year**
2. **Cost £60,000, £20,000 a year**
3. **Cost £40,000, £10,000 a year**

**Why:** Payback: 2.5, 3 and 4 years.

# Topic tasks

## 4 Payback in months

Round 2

A new valeting bay.

**Payback period, in months?**

| | |
|---|---|
| Cost to set up | **£17,100** |
| Returns, each year | **£8,290** |
| Running costs, each year | **£1,450** |

**ANSWER**

**30 months**

£8,290 − £1,450 = £6,840;
£17,100 ÷ £6,840 × 12 = 30

**Why:** It has paid for itself after 30 months.

## 5 Ramp or bay

Round 3

Ramp: payback 2 years, ARR 10%. Bay: payback 4 years, ARR 18%. Demand is stable; cash is fine.

**Which project should you choose, and why?**

**A** The ramp

**B** The bay

**ANSWER**

✔ **The bay**

**Why:** With little risk and cash to spare, the higher return wins.

## 6 Quicker payback

Round 3

Which pays back first?

**Payback of the quicker option, in months?**

| | |
|---|---|
| Option A: cost | **£17,700** |
| Option A: returns a year | **£7,080** |
| Option B: cost | **£38,880** |
| Option B: returns a year | **£12,960** |

**ANSWER**

**£30**

A 30 months; B 36 months; quicker 30

**Why:** The shorter payback gets the money back sooner.

# Final questions

Each pupil answers 6 final questions on investment appraisal. They come from these 11 questions.

1. An inspection line costs £300,000 and should make £45,000 average annual profit. On original cost, what is its accounting rate of return?
   ✔ **15%**

2. A £180,000 paint booth returns £60,000, £75,000 and £90,000 over three years. Ignoring timing within years, what is its payback period?
   ✔ **2.5 years**

3. A project requires £120,000 now and has discounted future cash inflows of £138,500. What is its net present value?
   ✔ **£18,500**

4. A project costs £200,000 and its discounted future cash inflows total £224,000. What is NPV?
   ✔ **£24,000**

5. A system costs £250,000 and is expected to earn average annual profit of £30,000. What is ARR using original cost?
   ✔ **12%**

6. An energy-saving project costs £180,000. Discounted savings are £205,000, but a downside scenario reduces those savings by 15%. What is NPV in the downside scenario?
   ✔ **−£5,750**

7. A £90,000 paint booth brings in £30,000 in year 1, then £40,000 a year. What is its payback period?
   ✔ **2.5 years**

8. A dealer spends £60,000 on a new workshop ramp, which brings a net cash inflow of £24,000 a year. What is the payback period?
   ✔ **2.5 years**

9. Two projects cost the same. Project A pays back in 2 years and project B in 4 years. Why might a cautious dealer prefer A?
   ✔ **It gets its money back sooner, so there is less risk**

10. A dealer invests £80,000 in a valeting centre. Over 5 years the total profit, after taking off the £80,000 cost, is £40,000. What is the ARR?
    ✔ **10%**

# Final questions

**11** Project X has an ARR of 14% and project Y an ARR of 9%. The dealer borrows at 11% a year. On ARR alone, what follows?

✔ **Only project X earns more than the cost of borrowing**